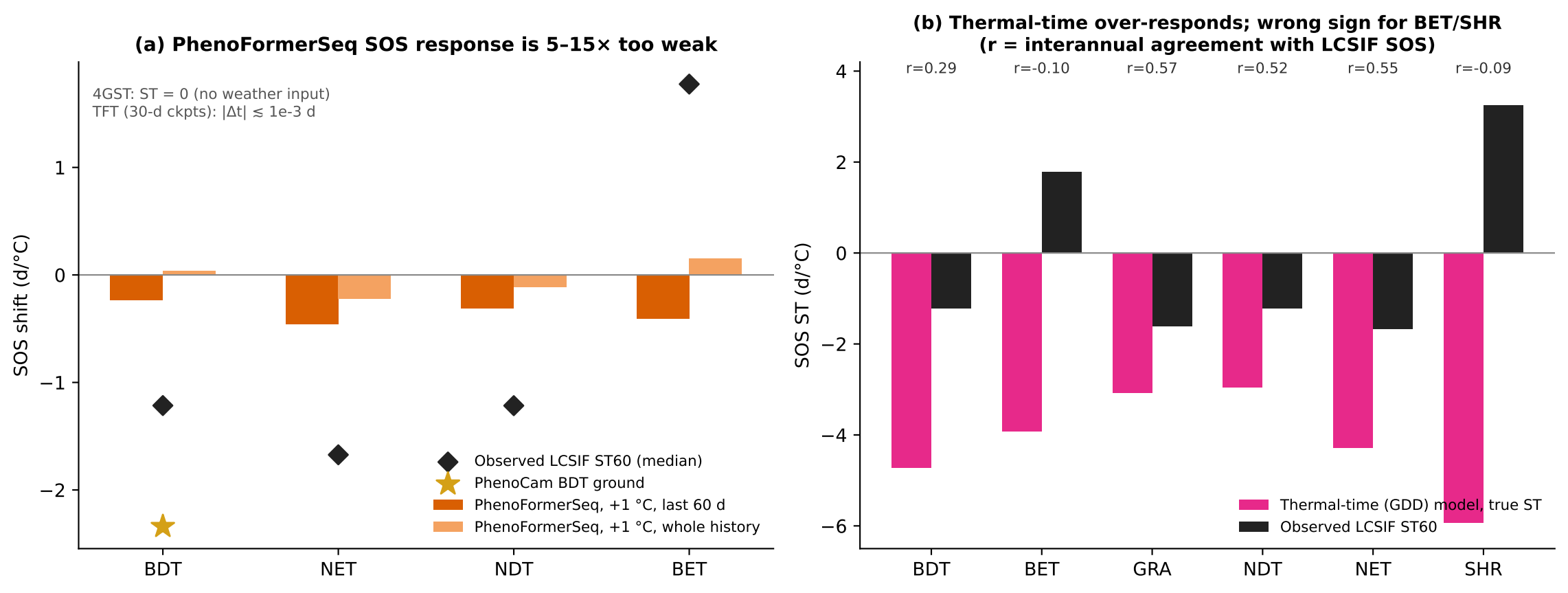
(a) PhenoFormerSeq SOS response is 5–15× too weak
4GST: ST = 0 (no weather input)
TFT (30-d ckpts): |Δt| ≲ 1e-3 d
SOS shift (d/°C)
1
0
−1
−2
BDT
NET
NDT
BET
Observed LCSIF ST60 (median)
PhenoCam BDT ground
PhenoFormerSeq, +1 °C, last 60 d
PhenoFormerSeq, +1 °C, whole history
(b) Thermal-time over-responds; wrong sign for BET/SHR
(r = interannual agreement with LCSIF SOS)
r=0.29
r=-0.10
r=0.57
r=0.52
r=0.55
r=-0.09
SOS ST (d/°C)
4
2
0
−2
−4
−6
BDT
BET
GRA
NDT
NET
SHR
Thermal-time (GDD) model, true ST
Observed LCSIF ST60